Clinical EEG and Neuroscience
2017, Vol. 48(1) 3–10
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DOI: 10.1177/1550059416645980
eeg.sagepub.com

Mismatch Negativity in First-Episode Schizophrenia: A Meta-Analysis

Sarah M. Haigh[1], Brian A. Coffman[1], and Dean F. Salisbury[1]

Abstract

Mismatch negativity (MMN) to deviant stimuli is robustly smaller in individuals with chronic schizophrenia compared with healthy controls (Cohen's $d > 1.0$ or more), leading to the possibility of MMN being used as a biomarker for schizophrenia. However, there is some debate in the literature as to whether MMN is reliably reduced in first-episode schizophrenia patients. For the biomarker to be used as a predictive marker for schizophrenia, it should be reduced in the majority of cases known to have the disease, particularly at disease onset. We conducted a meta-analysis on the fourteen studies that measured MMN to pitch or duration deviants in healthy controls and patients within 12 months of their first episode of schizophrenia. The overall effect size showed no MMN reduction in first-episode patients to pitch-deviants (Cohen's $d < 0.04$), and a small-to-medium reduction to duration-deviants (Cohen's $d = 0.47$). Together, this indicates that pitch-deviant MMN is not a candidate biomarker for schizophrenia prediction, while duration-deviant MMN may hold some promise, albeit nearly a third as large an effect as in chronic schizophrenia. Potential causes for discrepancies between studies are discussed.

Keywords

mismatch negativity, schizophrenia, first-episode, pitch, duration, effect size

Received September 29, 2015; revised March 23 2016; accepted March 31, 2016.

Introduction

Individuals with schizophrenia demonstrate sensory and cognitive deficits that appear to worsen after the first episode of psychosis. There are numerous reports of structural and functional neurological deficits that correlate with disease course,[1-7] highlighting the need for a biomarker that can help identify those who are at risk for developing schizophrenia. Identifying at-risk individuals before their first psychotic break offers the opportunity for early pharmacological and/or behavioral intervention, potentially preventing the postbreak neurological deficits.[8] Cognitive and psychosocial intervention at first episode has improved symptoms, functioning, and hospitalizations in these patients,[9,10] even 10 years after the intervention began,[11] and may have greater benefits prior to first-episode.

Mismatch negativity (MMN) is one potential electrophysiological biomarker for early identification. Biomarkers are defined to include, ". . . biological characteristics that can be objectively measured and evaluated as an indicator of normal biological processes, pathogenic processes or pharmacological responses to a therapeutic intervention."[12] We have previously argued that MMN is a potential biomarker for disease progression and as an outcome measure for pharmacological intervention.[7] However, many studies have suggested that MMN be a biomarker for the presence of the disease at the first break for schizophrenia. The purpose of this article is to assess this. MMN is an event-related potential (ERP) that appears with the presentation of deviant stimuli.[13] For example, single tones that differ in pitch or duration among repeated standard tones elicit an MMN. Source analysis of MMN in EEG and magnetoencephalography signal localizes MMN to auditory cortex,[14-16] with an additional later source from prefrontal cortex.[14,17]

MMN also has some clinical utility.[18,19] Individuals with chronic schizophrenia exhibit reduced MMN amplitudes, 0.94 SDs smaller for pitch deviants and 1.23 SDs smaller for duration deviants compared with healthy controls.[20] MMN correlates with reductions in gray matter,[7] correlates with disease severity and cognitive dysfunction,[21] and impaired social functioning.[22] MMN reductions may reflect deficits in NMDA receptors, [23-26] which are abnormal in schizophrenia,[27-30] suggesting a neurological mechanism for MMN that can be tracked through the progression of the disease. MMN deficits also appear to be more severe in schizophrenia compared to individuals with bipolar disorder[31] or Alzheimer's disease. [32]

[1]Western Psychiatric Institute and Clinic, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

Corresponding Author:
Sarah M. Haigh, Clinical Neurophysiology Research Laboratory, Western Psychiatric Institute and Clinic, Department of Psychiatry, University of Pittsburgh School of Medicine, 3501 Forbes Avenue, Suite 420 Oxford Building, Pittsburgh, PA 15212, USA.
Email: haighsm@upmc.edu